SWC1IC: Short-Range Wireless Communication – WS 2026
Lab Report 1

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Abstract

This Lab Unit covered the basics of setting up radio transmission on a Nordic nRF52840-DK development board. Through a hand on exercise the signal strength and packet loss between two boards was tested at various distances. This showcased both the effect of transmission strength and different radio modes.

1.1 The first assignment

The goal of this assignment is to analyze the wireless performance of the nRF52840 SoC by measuring signal strength (RSSI) and packet loss over different distances, transmit powers, and modulation schemes.

- **Setup and Implementation:** Two Nordic nRF52840-DK boards (TX and RX) were controlled via PuTTY over serial UART (115200 baud, 8N1) using the Zephyr `radio_test` sample. An interference-free channel at 2440 MHz was selected (`start_channel 40`) via the nRF Connect RSSI Viewer. The TX sent bursts of 1000 packets with `pattern_11110000` (`start_tx_modulated_carrier 1000`) at 0 dBm with BLE 2 Mbit/s. The RX logged received packets N_{RX} and RSSI with `print_rx`. Packet loss was calculated as:

$$\text{Packet Loss } [\%] = \frac{1000 - N_{\mathrm{RX}}}{1000} \times 100 \tag{1.1}$$

- **Test Procedure:** Measurements were performed across five physical distances: 1 m, 4 m, 6 m, 8 m, and 11 m.

Summary and Comments

- **RSSI vs. Distance (Figure 1.1):** At 1 m, the signal remained flat near −61 dBm. Increasing the distance lowered the RSSI to −72 dBm at 4 m, −85 dBm at 8 m, and down to −88 dBm at 11 m. Noticeable fluctuations appeared at larger distances due to indoor multipath reflections.

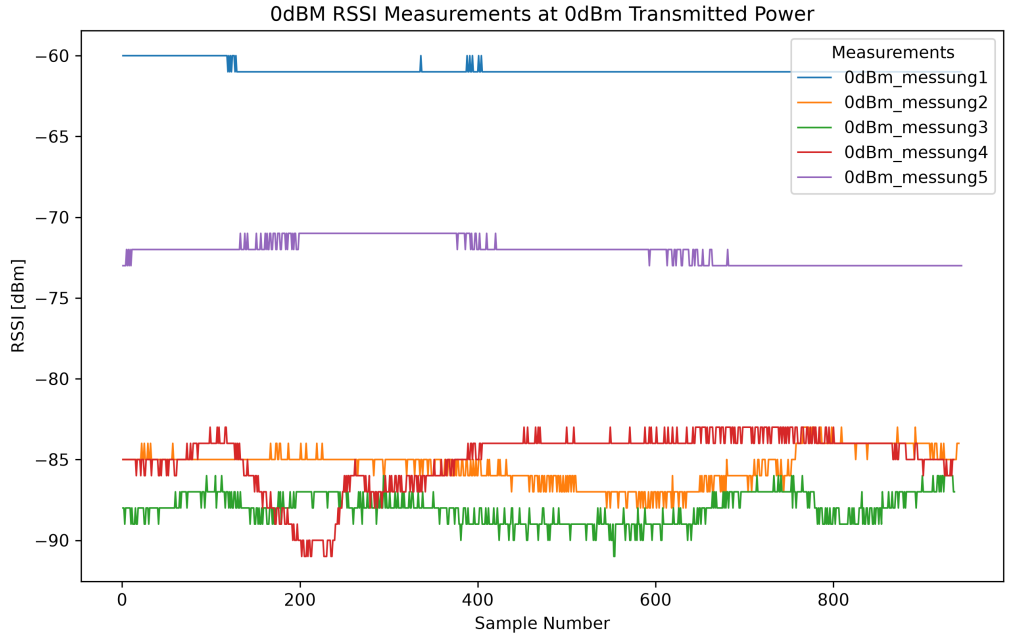


Figure 1.1: Received RSSI across 1000 samples for each distance at 0 dBm transmit power.

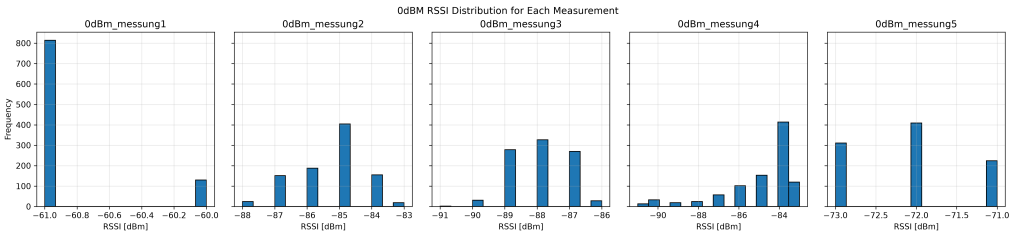


Figure 1.2: RSSI histogram distributions across all five test positions.

- **RSSI Spread (Figure 1.2):** Close distances (1 m and 4 m) exhibited narrow spreads within 1–2 dBm. Distances from 6 m to 11 m broadened across 5–8 dBm, demonstrating that attenuated signals are far more sensitive to environmental changes and room movement.
- **Packet Reception and Loss (Figure 1.3):** Out of 1000 transmitted packets, 945 were received at 1 m (5.5% loss) and 936 at 11 m (6.4% loss). Despite the 27 dBm signal drop, packet loss remained below 6.5%, showing that 0 dBm provides ample link margin for indoor spaces.

Key Takeaways

- **Indoor Attenuation:** Physical obstructions and multipath reflections prevent RSSI from decreasing linearly with distance, making raw RSSI unreliable for stan-

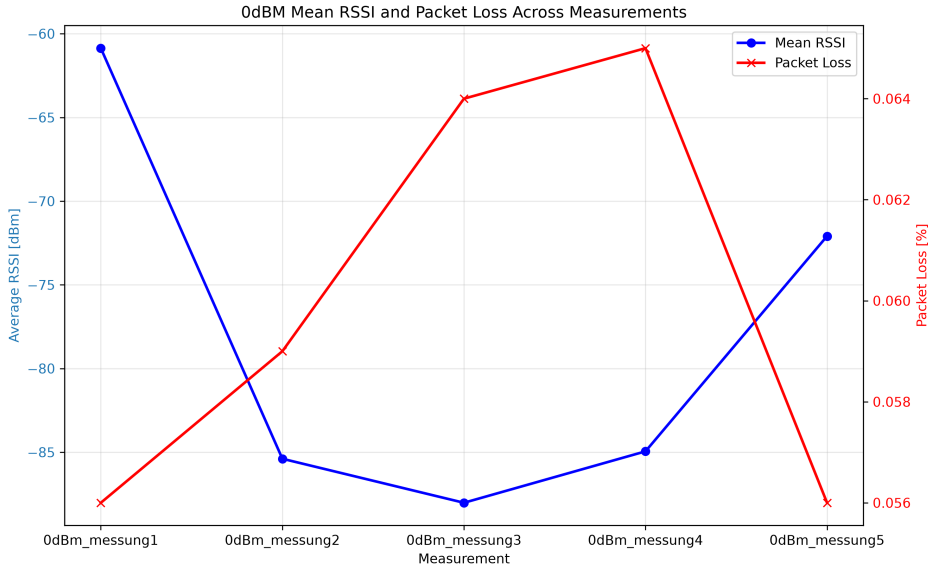


Figure 1.3: Mean RSSI compared with packet loss across all measured distances.

dalone distance estimation.

- **Link Margin at 0 dBm:** At 0 dBm, BLE 2 Mbit/s easily covers up to 11 m with negligible loss while the -40 dBm had considerably higher packet loss, indicating that higher transmit power is crucial for maintaining reliable links in indoor environments.